

Enterprise Infrastructure Optimization & High Availability Automation Framework

Titan Award Submission – Enterprise Innovation in Healthcare Data Platforms

Enterprise Challenge Landscape

Enterprise Scale

- Multi-terabyte production databases
- Indexes exceeding 800–900 GB
- Mission-critical Medicaid systems

Operational Challenges

- Maintenance exceeded enterprise operational windows
- Severe fragmentation accumulation
- Transaction-log growth instability

HA / Stability Risks

- Always-On synchronization instability
- Replica recovery overhead
- Nightly batch disruption risk

Innovation Requirement

- Synchronization-aware orchestration
- Controlled parallel execution
- Intelligent log-threshold governance

Large-scale healthcare database environments required an intelligent, synchronization-aware optimization framework to safely manage enterprise index maintenance within high-availability operational constraints.

Hybrid Cloud & Platform Applicability

Platform Independence

- Designed for enterprise-scale database environments
- Applicable across on-premises, cloud, and hybrid deployments
- High-availability aware by design

Core Design Principles

- Synchronization-aware orchestration
- Intelligent operational governance
- Controlled execution management
- Automated recovery mechanisms
- Fault-tolerant processing

Enterprise Engineering Characteristics

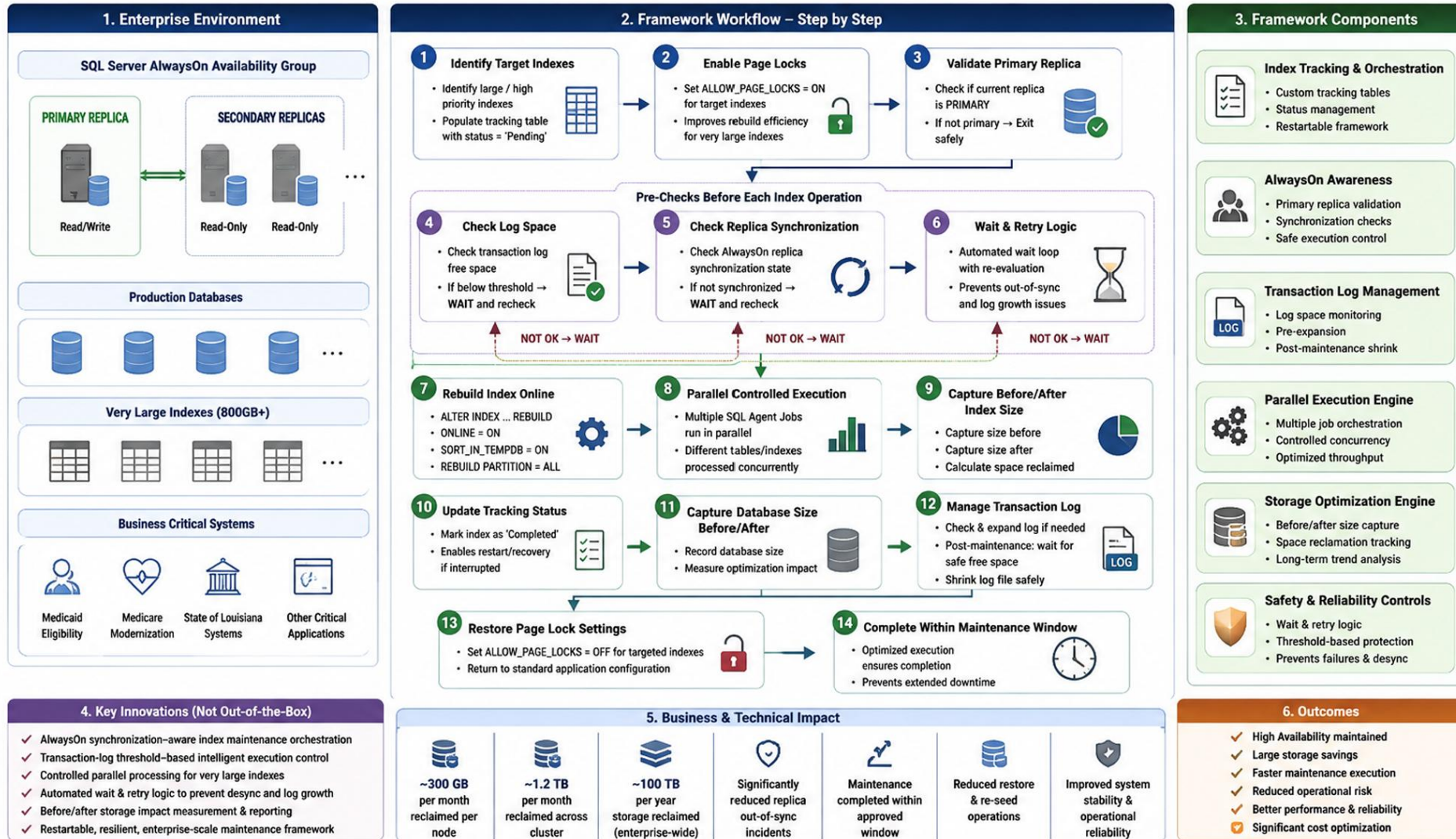
- Scalable architecture
- Operational resiliency
- Maintenance-window optimization
- Governance-driven execution
- Production-ready automation

The framework extends beyond traditional maintenance approaches by introducing an intelligent orchestration layer that continuously evaluates operational conditions, platform readiness, and governance controls before executing optimization activities.

Optimization Framework Architecture

Enterprise Database Optimization & High Availability Automation Framework

AlwaysOn–Aware Index Maintenance with Log Management & Storage Optimization

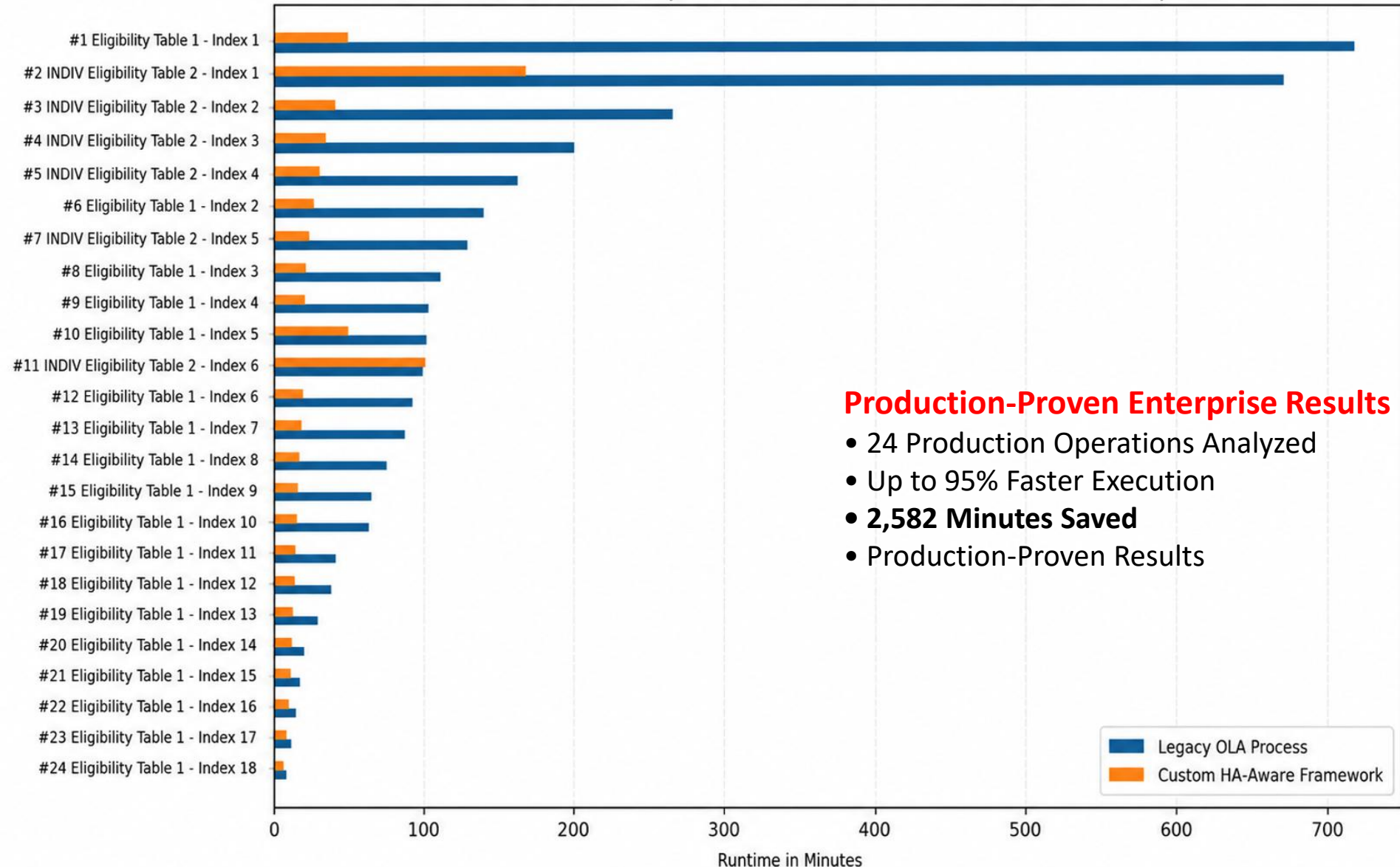


Custom-Built Innovation: This end-to-end optimization framework integrates Microsoft SQL Server features in a unique way to solve enterprise challenges at scale. There is no single out-of-the-box Microsoft feature that combines all these capabilities together for AlwaysOn environments.

Runtime Benchmark Comparison

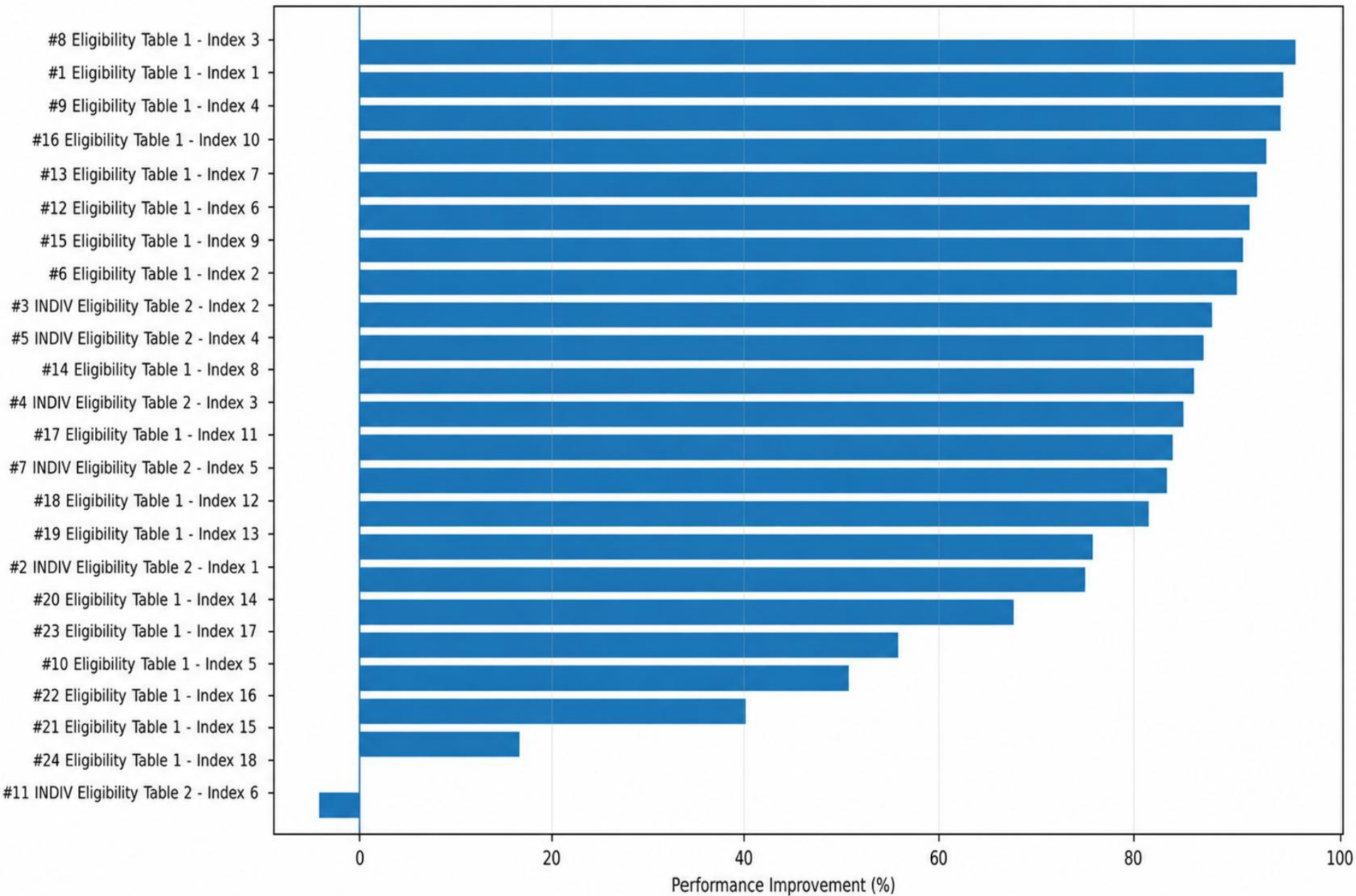
Up to 95% runtime reduction across production index maintenance operations

Before vs After Runtime Comparison - 24 Production Index Maintenance Operations



Performance Improvement Metrics

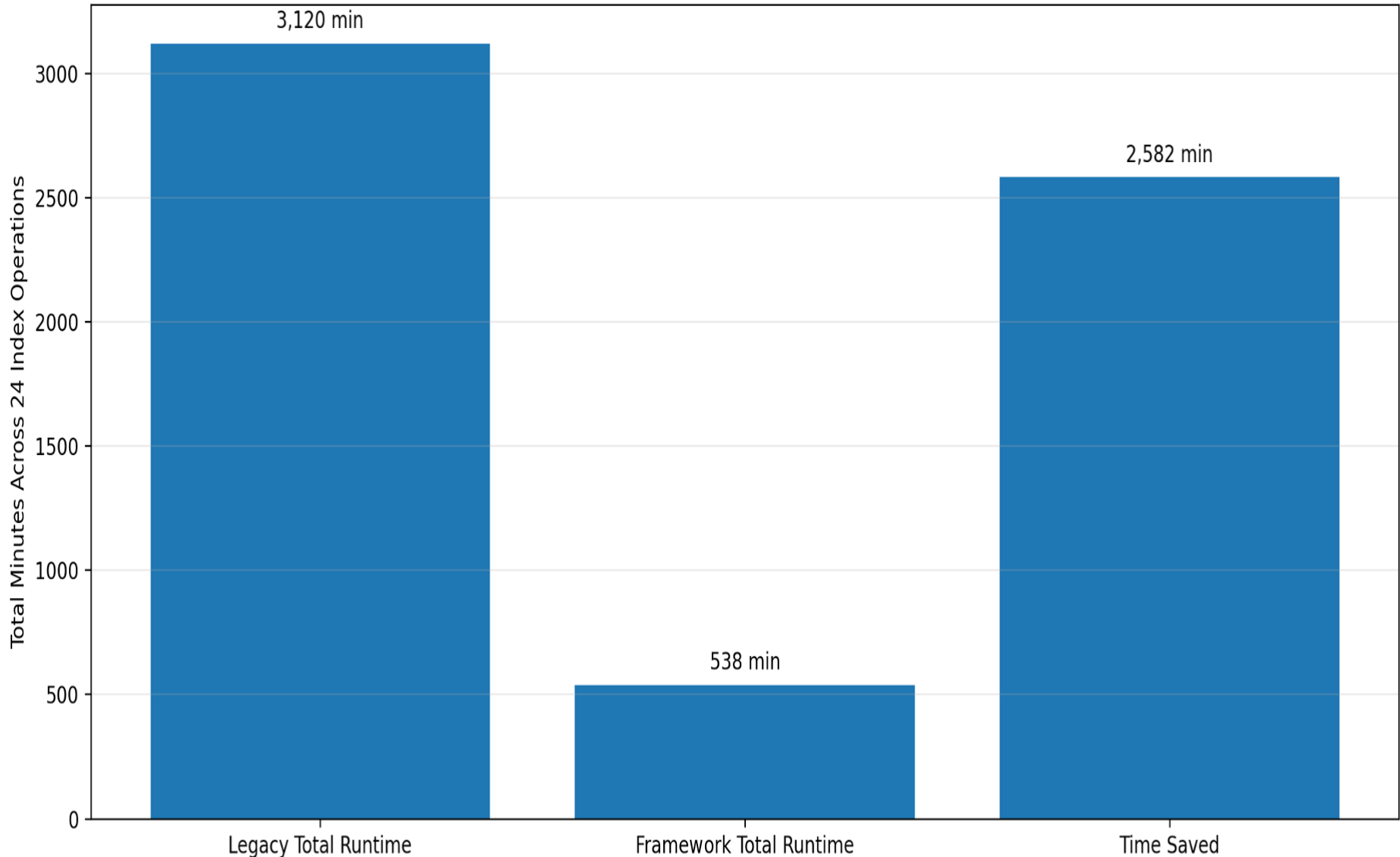
Performance Improvement by Index - Custom Framework vs Legacy Process



Runtime Reduction

2,582 minutes saved across 24 production operations

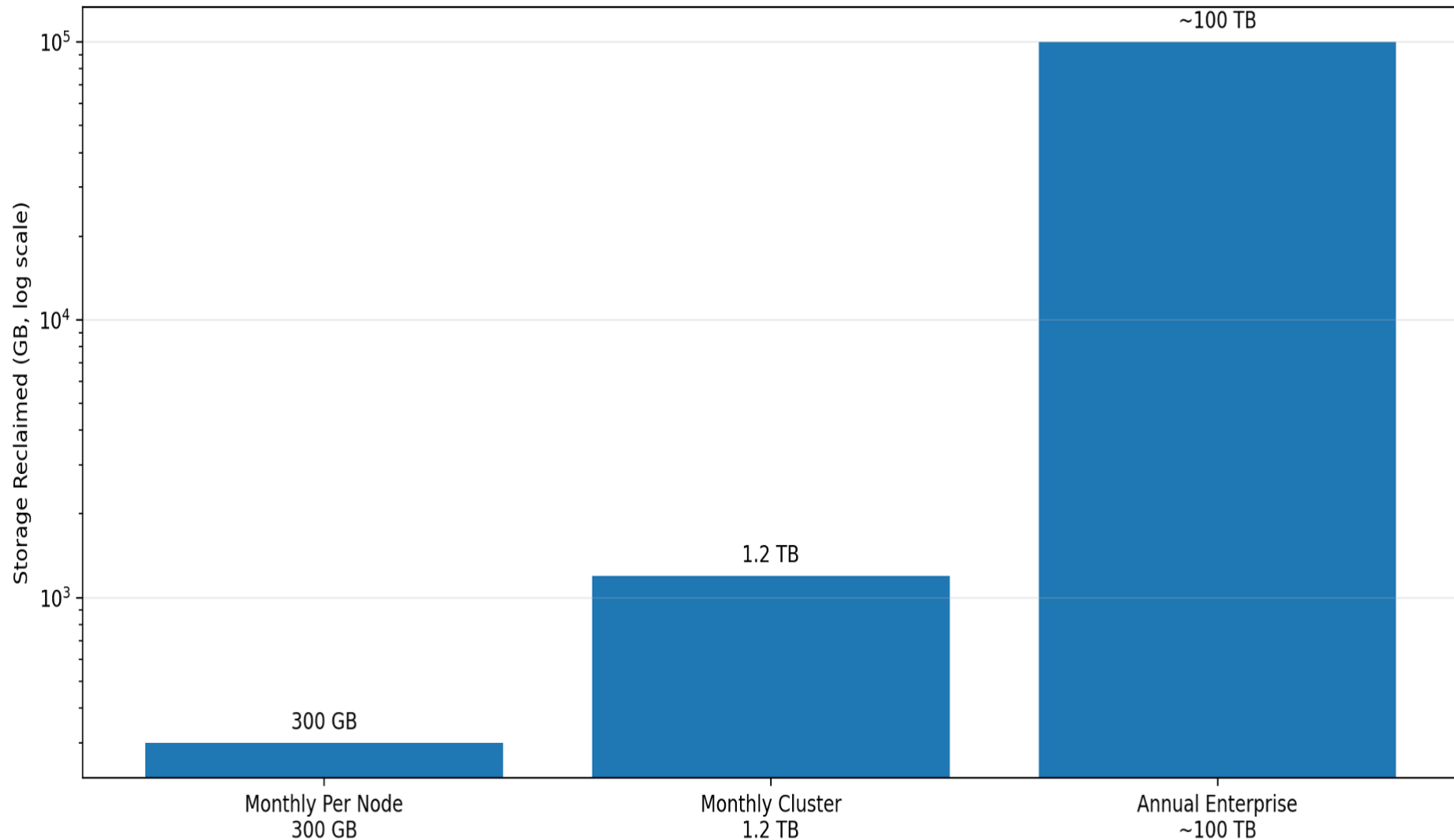
Aggregate Runtime Reduction Across Production Index Maintenance Set



Enterprise Storage Optimization

Over 100 TB annual enterprise storage optimization

Storage Optimization Impact - Production Cluster Reclamation



Detailed Benchmark Matrix - 24 Production Index Operations

This matrix can be used as supporting evidence in the award package. Table and index names have been anonymized to protect client and production system confidentiality. The matrix includes measured operational records with legacy duration, optimized framework duration, time saved, and percent improvement.

#	Table	Index	Legacy Min	Framework Min	Saved Min	Improvement
1	Eligibility Table 1	Index 1	708	41	667	94.2%
2	INDIV Eligibility Table 2	Index 1	667	155	512	76.8%
3	INDIV Eligibility Table 2	Index 2	262	31	231	88.2%
4	INDIV Eligibility Table 2	Index 3	193	28	165	85.5%
5	INDIV Eligibility Table 2	Index 4	151	19	132	87.4%
6	Eligibility Table 1	Index 2	128	12	116	90.6%
7	INDIV Eligibility Table 2	Index 5	123	20	103	83.7%
8	Eligibility Table 1	Index 3	107	5	102	95.3%
9	Eligibility Table 1	Index 4	100	6	94	94.0%
10	Eligibility Table 1	Index 5	97	47	50	51.5%
11	INDIV Eligibility Table 2	Index 6	94	97	-3	-3.2%
12	Eligibility Table 1	Index 6	88	7	81	92.0%
13	Eligibility Table 1	Index 7	83	6	77	92.8%
14	Eligibility Table 1	Index 8	67	9	58	86.6%
15	Eligibility Table 1	Index 9	57	5	52	91.2%
16	Eligibility Table 1	Index 10	57	4	53	93.0%
17	Eligibility Table 1	Index 11	31	5	26	83.9%
18	Eligibility Table 1	Index 12	30	5	25	83.3%
19	Eligibility Table 1	Index 13	26	6	20	76.9%
20	Eligibility Table 1	Index 14	15	5	10	66.7%
21	Eligibility Table 1	Index 15	12	10	2	16.7%
22	Eligibility Table 1	Index 16	10	6	4	40.0%
23	Eligibility Table 1	Index 17	9	4	5	55.6%
24	Eligibility Table 1	Index 18	5	5	0	0.0%

Measured Business & Operational Impact

Performance Transformation

708 min



41 min

94% Faster

107 min



5 min

95% Faster

100 min



6 min

94% Faster

- Reduced maintenance-window failures
- Improved AlwaysOn operational stability
- Production-proven optimization framework

Enterprise Business Value

300 GB

Monthly reclaimed
per node

1.2 TB

Monthly reclaimed
across cluster

24x7

Operational
stability
improvement

HA

High-availability
optimization

The framework transformed enterprise maintenance execution into a synchronization-aware optimization platform capable of safely operating within high-availability healthcare environments.

Enterprise Innovation Outcomes

Innovation Contributions

- Architected the enterprise optimization framework
- Led synchronization-aware orchestration design
- Directed HA optimization strategy
- Designed transaction-log governance controls
- Improved maintenance-window compliance
- Delivered enterprise operational stability

Enterprise Outcome

- Up to 95% runtime reduction
- Improved high-availability reliability
- ~1.2 TB monthly storage reclaimed
- Reduced operational recovery overhead

Innovation Highlights

Parallel

Controlled parallel execution

HA-Aware

Sync-aware maintenance

Log Control

Intelligent log governance

Enterprise

Production-proven impact

KPI Metrics

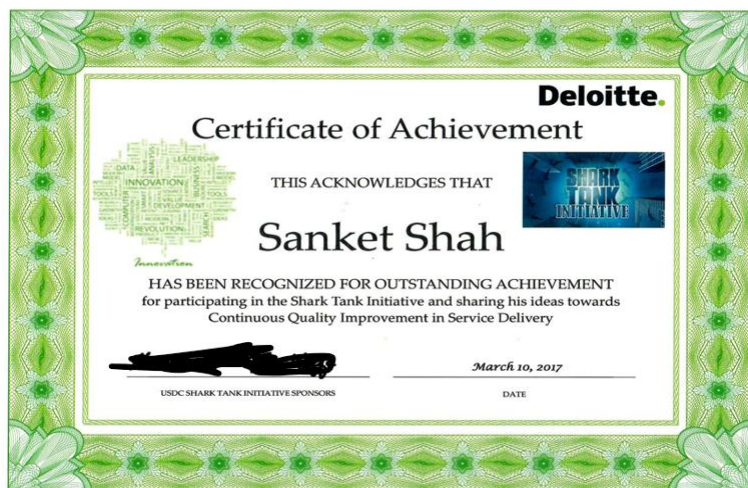
- Indexes exceeding 800–900 GB
- 1.2 TB monthly reclaimed
- 94–95% faster execution
- Multi-terabyte production environment

The framework delivered measurable enterprise-scale operational improvements and received internal organizational recognition for innovation, infrastructure optimization, and high-availability excellence.

Industry & Organizational Recognition

- Internal innovation recognition through Deloitte Shark Tank Initiative
- Recognition for operational excellence in High Availability and infrastructure optimization
- Applied within mission-critical healthcare modernization environments
- Demonstrated measurable enterprise-scale reliability, performance, and operational improvements

Supporting Recognition & Achievement Awards



Deloitte.



Congratulations Sanket,

Congratulations and thank you for your performance that demonstrates your dedication and willingness to go the extra mile!

You have been chosen to receive an **Applause Award** in the amount of \$200, in acknowledgement of your contributions to the success of the U.S. firms. Specifically, you are being recognized for the following contribution:

Thank you for resolving the SQL HA issues. Keep up the good work.

Thank you for your continued commitment to Deloitte, and we wish you much future success!

Applause Award is presented to:

Sanket Shah

06/28/17

These recognitions provide independent validation of the innovation mindset, operational excellence, and infrastructure optimization expertise that contributed to the development of this framework.

Conclusion

The custom enterprise optimization framework transformed traditional database maintenance into a synchronization-aware automation platform engineered for mission-critical healthcare modernization environments.

Framework

- Synchronization-aware orchestration
- Controlled parallel index execution
- Intelligent transaction-log governance
- Enterprise-scale automation framework
- Beyond traditional enterprise maintenance capabilities

Enterprise Impact

- Utilized across SDLC, SIT, UAT, training, and production environments
- Supported load testing and database split proof-of-concepts
- Enabled stable maintenance of 800–900 GB indexes
- Reclaimed over 100 TB annually enterprise-wide
- Improved AlwaysOn operational reliability

The implementation demonstrated enterprise-scale innovation, platform engineering leadership, and production-proven optimization impact within high-availability healthcare data systems.

The integrated orchestration, synchronization-aware execution model, and intelligent transaction-log governance introduced through this custom framework delivered capabilities beyond traditional maintenance solutions, including intelligent governance, operational resiliency, and high-availability-aware automation.